YMnotes

CHAPTER 07

Foodborne Illness and Food Poisoning

Updated edition · built on the 2026–2027 handout, with notes from the earlier YMnotes chapter and a supplement

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HOW TO READ THE COLOURS

- **Green** · the 2026–2027 handout — the main line of this chapter.
- **Blue** · **From the earlier edition:** a detail found only in the older chapter.
- **Amber** · **Clinical clarification:** a source statement checked against current guidance.
- **Red** · points the handout itself highlighted (boxed or in red/blue).
- **Brown note** · editorial note: spelling, layout or an unclear source line.
- **Slate** · supplement from the earlier edition only; its presence does not prove it is in the current exam.

FROM THE EARLIER EDITION

LEARNING OBJECTIVES

- Explain the terminology related to gastrointestinal infection.
- Discuss the etiology, pathogenesis, epidemiology and prevention of microbial infection of gastrointestinal infections.
- Discuss virulence mechanisms and correlate virulence mechanisms with specific pathogens.
- Describe the laboratory diagnosis for bacteria and viruses.
- Discuss different methods for preventing foodborne illness and food poisoning.

PART 1

1 Foodborne illness

Salmonella: poultry are considered the primary reservoirs of the organism causing:

Salmonellosis or enteric fever (typhoid)

Resulting from **bacterial invasion of the bloodstream**.

Acute gastroenteritis

Resulting from a food-borne infection mainly contaminated **eggs**, and intoxication.

EDITORIAL NOTE

The handout's outline line reads "a. Enteric fever · b. *Helicobacter pylori–Campylobacter pylori*". The handout text contains no separate *H. pylori* section; *Campylobacter* is covered in section 1.4 and enteric fever in 1.2.

FROM THE EARLIER EDITION

Foodborne illness is described as infections caused by a variety of microorganisms, mainly *Salmonella*, *Clostridium perfringens*, *Campylobacter* and *Staphylococcus aureus*; **Norovirus** is also listed.

Salmonella is a member of the **Enterobacteriaceae** family.

Salmonella — the organism

SALMONELLA · MICROBIOLOGY AT A GLANCE

Morphology	Gram-negative bacilli, **motile**, non-spore forming.
Culture	**Facultative anaerobes**; grow well at 37 °C after 24 hours on **MacConkey agar**: **non-lactose fermenters** (pale yellow colonies). Also **SS agar**, **deoxycholate citrate agar (DCA)** and enrichment broth media, which include **selenite** and **tetrathionate**.
Biochemical reactions	**Partial** sugar fermentation, with production of acid and gas by most species **except *S. typhi*, which produces acid only**. Urease **negative**, oxidase **negative**, H_2S **positive**.
Antigenic structure	**O (somatic) antigens** and **H antigen (flagella) antigens**: used to determine the **type** within the groups. **Virulent antigen (Vi Ag)**: anti-phagocytic action, present **only in *S. typhi***. **Earlier edition:** O antigens subdivide the *Salmonellae* into **groups**; H antigens determine the **type** within the groups.
Pathogenesis	Due to **endotoxins and exotoxins** (e.g. enterotoxins causing diarrhea and body-fluid loss) and antigens.
Resistance	Killed by heating at **60 °C for 30 min**; antibiotics and other chemotherapy agents are effective. Resistant to the action of **bile salts**.

A Enteric fever

Caused by *S. typhi*; the milder **paratyphoid fever** is caused by *Paratyphi* A, B and C. Enteric fever is defined as a **generalized infection of the reticuloendothelial system and lymphoid tissue accompanied by sustained fever and bacteremia**. Enteric fevers are **endemic in Egypt**.

Pathogenesis and clinical picture

Enteric fever is **a strictly human disease**, from case or carrier stool and urine. The organisms enter the human digestive tract with contaminated food or drink, e.g. **raw shellfish, milk products and raw vegetables**.

PATHWAY OF *SALMONELLA TYPHI* IN THE BODY — IN THE HANDOUT'S ORDER

1. **Adhere** to the mucosa of the **small intestine by pili**
2. **Penetrate** to the submucosal layer, **causing no lesion**
3. **Invade macrophages** in lymphoid tissue; bacterial multiplication occurs
4. **Spread** transported to **mesenteric lymph nodes** and lymphatics, then released into the **bloodstream**
5. **Seed organs** carried to many organs; replication inside macrophages occurs — bone marrow; kidneys; intestine; spleen; liver
6. **Septicemia** after multiplication in the reticuloendothelial system it is released into the bloodstream → signs of septicemia, **mainly fever**
7. **Gall bladder** the organism is transported there to be **excreted with bile**
8. **Stool** Salmonella multiply in intestinal lymphoid tissue and are **excreted in stools at the second week**

FROM THE EARLIER EDITION

Salmonella organisms **survive intracellularly and replicate in macrophages** due to **decreased fusion of lysosomes with phagosomes**, as well as the presence of surface proteins (**defensins**) which allow the bacteria to withstand oxygen-dependent and oxygen-independent intracellular killing. When released from macrophages, the **Vi capsular antigen** (only in *S. typhi*) withstands **complement-mediated killing**.

Earlier-edition wording kept as printed ("surface proteins defensins").

Clinical course

Incubation	Fever	Organs and blood	After cure
10–14 days	Gradually elevated in the **first week**, then **steady** in the second week.	**Enlargement of spleen and liver.** White cell count **normal or low**.	Salmonella excretion by patients **may continue long after clinical cure**.

Outcome: 5% of patients cured from typhoid **remain carriers** for months or even years. **Urinary carriers are common in Egypt because of bilharziasis.**

Laboratory diagnosis

Specimens are collected **according to the stages** of the disease.

FIRST WEEK	SECOND WEEK	SEROLOGY · WIDAL
Blood culture is done in **fluid media**, then subcultured on solid media, mainly **MacConkey's** medium at 37 °C.	*Salmonella* is isolated **from stool** by direct culture on broth and solid differential selective media, **then from urine** by direct culture on solid media.	Antibodies **rise sharply during the second and third weeks** of enteric fevers.

FROM THE EARLIER EDITION

Isolation is listed as: (A) first week — blood culture; (B) second week — stool, "**and after the second week from urine**", by direct culture on selective media. The updated handout places urine culture in the second week, after stool.

Serological diagnosis by the Widal agglutination test

This test demonstrates **agglutination** of the patient's serum antibodies against the **O (somatic) antigens** of *Salmonella* and the **H (flagellar) antigens** of *S. typhi* and *Paratyphi* A, B, C. Demonstration of **rising titers** in **paired samples** indicates acute recent infection. The results should include the titer of **both O and H antibodies and Vi antibodies**. The presence of **IgM** is indicative of acute recent infection.

1 **O antigen**
High or rising titer of antibody to **O antigens (≥ 1:160)** suggests that **active infection** is present.

2 **H antigen**
High titer of antibody to **one type of H antigen (≥ 1:160)** suggests **recent infection** with this type of *Salmonella*.

3 **Vi antigen**
High titer of antibody to the **Vi antigen** occurs in some **carriers**.

4 **O + several H**
High titers of antibodies to O antigens and **more than one H antigen** indicate **recent immunization**. With old immunization or past infection, **O antibodies disappear** and only antibodies to **H antigens** remain.

HIGHLIGHTED IN THE HANDOUT

The Widal test has many limitations as a diagnostic serological test for enteric fever. Several factors need to be considered for interpretation:

1. **Endemicity of enteric fever** in the area of investigation. Healthy persons have antibody titer due to subclinical infection. **Titers up to 1:160 are considered normal in Egypt. False-positive** results may occur in an endemic area, with immunization with any typhoid vaccine, with a previous infection, or with infection by other related members of *Enterobacteriaceae*.
2. **False negative** in case of administration of **antibiotics**, and if the Widal test is done in the **early stage** of disease, where low antibody titers are found.

Antibiotics may not be effective on *Salmonella* carriage because the site of carriage may not allow penetration by the antibiotic, e.g. due to the presence of fibrous tissue.

These lines appear twice on the handout's pages 2–3; they are given once here.

Immunity

Persistent immunity is acquired after enteric disease. **CMI** (cell-mediated immunity) is the main immunity. **Humoral immunity destroys the organism in the blood.** Antibody titer can continue for a long time after recovery.

Prevention and control — immunization

Four types of typhoid vaccines are currently available for use:

#	Vaccine	Details given in the handout
a	**Parenteral heat-phenol-inactivated vaccine** (TAB)	Contains *S. typhi* and *paratyphi* A, B, C — hence the name **TAB vaccine**. Given as **2 subcutaneous (SC) doses. This is the current vaccine used in Egypt.**
b	**Acetone-inactivated** parenteral vaccine	—
c	**Oral live-attenuated** vaccine	For *S. typhi*. **Should not be given to children younger than 6 years of age.**
d	**Vi capsular polysaccharide (ViCPS)**	A newly licensed capsular polysaccharide vaccine for parenteral (**intramuscular**) use.

No typhoid vaccine is 100% effective and is not a substitute for good hygiene.

CLINICAL CLARIFICATION

The vaccine statements above ("current vaccine used in Egypt", ViCPS "newly licensed") are the handout's and are kept for study. The current WHO typhoid fact sheet describes **typhoid conjugate vaccines** (two prequalified since December 2017) and says the older vaccines do not give long-lasting immunity. Always follow current national guidance in practice.

Source: WHO, Typhoid fact sheet (who.int).

B Salmonella food poisoning

Salmonella typhimurium — food-borne toxic infection

About **12–24 hours** after ingestion of contaminated food with a **large number of organisms**, symptoms appear (**diarrhea, vomiting, fever**) and last **2–5 days**. **Spontaneous cure** usually occurs. Contamination of **meat with rat excreta** most often occurs.

Salmonella enteritidis — egg-associated salmonellosis

The egg can be **inside perfectly normal**. Incubation period **12 to 72 hours** after consuming a contaminated food. The illness usually lasts **4 to 7 days**, and most persons **recover without antibiotic treatment**.

FROM THE EARLIER EDITION

For *S. typhimurium* the earlier chapter words it as: symptoms appear after 12–24 hours following ingestion of **meat contaminated with rat excreta which contain a large number of organisms**; it lasts 2–5 days.

PART 1 · CONTINUED

1.4 *Campylobacter*

12–15 species are associated with human disease:

Species	Source named in the handout
Campylobacter jejuni	poultry; humans
Campylobacter coli	porcine; humans
Campylobacter fetus	cattle; sheep

MICROBIOLOGY

Morphology	Gram-negative **curved bacilli** (**S-shaped** or "**gull-winged**" appearance), non-spore forming, with a characteristic **rapid darting motility** by means of a long sheathed polar flagellum at one (**polar**) or both (**bipolar**) ends. Motility slows quickly in wet-mount microscopy preparations due to **aerotoxicity (oxygen sensitivity)**.
Culture	**Micro-aerophilic, thermophilic and capnophilic** (CO_2 10%), on selective media with added antibiotics such as **Skirrow's medium at 42 °C**, which is the natural body temperature of avian species.
Biochemical reactions	**Oxidase positive**, **catalase positive**, do not utilize carbohydrates.
Virulence factors	**Cellular components:** endotoxin (LPS), flagellum, adhesins and invasins. **Extracellular components:** enterotoxins (LT) and cytotoxins.

Transmission and clinical picture

Transmission: feco-oral route. *Campylobacter* is usually acquired from **food (specially poultry)**, **milk**, or contact with infected animals (e.g. dogs). **Children under 5 years and young adults** are the most frequently affected age groups.

Incubation period of 1–7 days with acute onset, **abdominal cramps, nausea but not vomiting**. The bacteria damage the mucosal surfaces of the **jejunum, ileum and colon**. Stools may contain **mucus or gross blood**. Generally self-limiting; symptoms may last for a week or longer. **Bacteremia is common** and *Campylobacter* is isolated from blood.

> IMPORTANT LATE-ONSET COMPLICATIONS FOLLOWING GASTROENTERITIS
>
> **Guillain-Barré syndrome · Reiter's syndrome · reactive arthritis · spontaneous abortion**

Laboratory diagnosis

Specimen collection and processing: feces should be examined **within a few hours**. Care is needed to **avoid oxygen exposure** while taking the sample — or, better, a **special transport medium** is used.

- **Direct examination:** Gram-stained smears of stools show **S-shaped bacilli**, and **darting motility** in wet-mount preparation under **dark-field illumination**.
- **Culture** (see Culture above). **Earlier edition:** culture is on **Skirrow's medium**.

Antibiotic resistance: fluoroquinolone resistance has developed rapidly, related to unrestricted use and the use of **enrofloxacin in poultry**. The organism is **inherently resistant to trimethoprim and most cephalosporins**.

PART 1 · CONTINUED

1.5 *Brucella* and brucellosis

Brucella is the cause of **brucellosis, a zoonotic disease**. It is a virulent, highly infectious bacterium and a **facultative intracellular pathogen** surviving in macrophages.

SPECIES NAMED IN THE HANDOUT

Species	Host
Brucella abortus	cattle
Brucella melitensis	goats, sheep
Brucella suis	swine, pigs
Brucella canis	dogs

Species are differentiated by **production of urease and H_2S**, dye sensitivity, cell-wall antigens and phage sensitivity.

MICROBIOLOGY

Morphology	Gram-negative **small bacilli**, non-spore forming, **non-motile** and **non-capsulated**.
Culture	**Aerobic**, **10% CO_2**; require **enriched media containing serum**, e.g. serum dextrose agar. Colonies appear after **48 to 72 hours**.
Biochemical reactions	Positive for **catalase, oxidase, urease**, and **reduce nitrates**.

Brucellosis (Malta fever or undulant fever): a **true zoonotic disease**, as **human-to-human transmission has not been identified**. The disease is characterized by **acute undulating fever**, i.e. bouts of fever for 3–4 weeks, then 3–4 weeks without fever.

> EDITORIAL NOTE
>
> "*B. melintensis*" in the handout is spelled here as *B. melitensis*.

Transmission

- **Direct contact** through skin abrasions or mucous membranes with infected animals or animal products (farmers, butchers and veterinarians).
- Consumption of **unpasteurized milk products and meat**.
- **Inhalation of aerosols** from infected animals, or laboratory-associated.

Laboratory diagnosis

Specimen: blood is taken during acute disease. Other samples according to the site of infection, such as joint fluid, CSF, pus. **Gram stain is of no significance.**

Method	Details
A · Culture	**Blood culture** is the method of choice early in the disease. Cultures need to be incubated for **up to 4–6 weeks** as they are slow-growing. It is important to **differentiate *Brucella* from *Salmonella* by urease test**: *Brucella* produces urease (**+ve**) while *Salmonella* does not (**–ve**).
B · Serology	The **main method** of laboratory diagnosis. **Two serum samples are collected 14–21 days apart.** A wide range of serum dilutions must be done in order to avoid a **false-negative result due to the prozone effect**, which is commonly present in cases of brucellosis as large amounts of antibodies are produced. Dilutions are done from **1:20 to 1:520**. **ELISA** is used to differentiate between specific **IgM and IgG** antibodies.
Brucellin test	An **intradermal test** based on **delayed-type hypersensitivity**. Interpretation of the result is difficult in persons frequently exposed to infection.

EDITORIAL NOTE

The handout labels the brucellin test "D" although no "C" appears; the dilution range "1:20 to 1:520" is printed identically in both editions and is kept as written — check it against your lab manual.

Prevention

Wearing **protective impermeable clothing, rubber boots, gloves and face masks** when exposed to animals. **Pasteurization of milk.** Treatment of domestic animals and **live attenuated vaccines for animals (cattle)**.

Treatment

Humans are treated with **combinations of antibiotics** for **4 to 6 weeks**.

PART 2

2 Food-borne poisoning

2.1 *Staphylococcus aureus*

S. aureus are **catalase-positive Gram-positive cocci**.

The toxin

Staphylococcal **enterotoxins do not change the taste, color or odor of food**. They are **heat stable** for 30 minutes of boiling and are **not inactivated by brief cooking**.

The source

A **food handler** carrying the organism in the **nose, hands**, or having superficial lesions such as a boil (abscess).

Common foods

Carbohydrate-rich food, e.g. cake, koshari, and **protein-rich** food, e.g. milk and its products.

Incubation period is short, 1–6 hours, due to consumption of **preformed enterotoxins**. The food poisoning is characterized by **vomiting and watery non-bloody diarrhea**.

Laboratory diagnosis

- **Specimen:** food, vomitus.
- **Microscopic examination:** Gram-positive cocci in **grape-like clusters**.
- **Culture: blood agar** — complete (beta) hemolysis — and **mannitol-salt agar**.

- **Typing** of *S. aureus* is done for **epidemiologic purposes**, in food-poisoning outbreaks. This could be achieved by **phage typing and PCR**.

> CROSS-REFERENCE · SECTION 2.2 OF THE HANDOUT
>
> **Salmonella food poisoning** — covered under *Salmonella* (see section 1.3).

2.2 Spore-forming Gram-positive bacilli

The Gram-positive spore-forming bacilli are the ***Bacillus*** and ***Clostridium*** species. ***Bacillus*** **species are aerobes** while **clostridia are anaerobes**.

Bacillus cereus — two forms of food poisoning

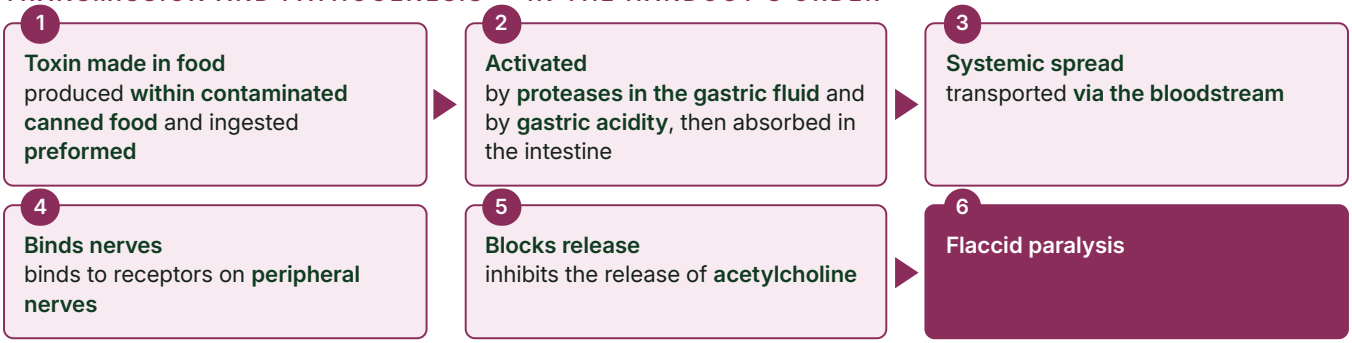
	a · "Short-incubation" (emetic) form	b · "Long-incubation" (diarrheal) form
Mediated by	**Preformed heat-stable enterotoxin**	**Heat-labile enterotoxin**. This exotoxin **activates adenylate cyclase**, increases **cAMP**, causing **profuse fluid secretion**.
Typical food	Usually in **improperly stored, cooked rice**	Present in **meat or vegetables**
Incubation	**1–6 hours**	**8–16 hours**
Note	Resembles *S. aureus* food poisoning	—

In either type, the illness usually lasts for less than 24 hours after onset.

2.3 *Clostridium botulinum* — botulism

Clostridium botulinum is the causative agent of **botulism**, a **rare but fatal** form of food poisoning, caused by a **potent botulinum toxin**.

TRANSMISSION AND PATHOGENESIS — IN THE HANDOUT'S ORDER



Botulinum toxin is rapidly acting; signs and symptoms appear **within 12 hours after ingestion**. As printed in the handout: "Antitoxin is only effective before the toxin binds the neuromuscular junction within 12 h after ingestion. Prognosis is poor for patients diagnosed after this time period."

CLINICAL CLARIFICATION · ANTITOXIN TIMING

The handout's "only effective within 12 hours" is not current guidance and should not be taken as a cut-off.

- CDC's 2021 clinical guidelines state that patients with suspected botulism **whose symptoms or signs (e.g. paralysis) are progressing should be treated with antitoxin regardless of the time that has elapsed since symptom onset**.
- The **greatest benefit goes to those treated within the first 2 days** of illness onset — so treat as early as possible, and **do not wait for laboratory confirmation** once clinical assessment supports botulism.
- Antitoxin **does not reverse paralysis that is already present**; it **arrests progression** by neutralizing toxin still circulating in the blood. Recovery of existing paralysis is slow and depends on supportive care (e.g. ventilation).

For the exam, know what the handout states; in clinical thinking, **early antitoxin is better, and later antitoxin can still be indicated while paralysis is progressing**.

Sources: CDC, Clinical Guidelines for Diagnosis and Treatment of Botulism, 2021 (MMWR Recomm Rep 70[2]); CDC Clinical Overview of Botulism; WHO Botulism fact sheet.

CLINICAL CLARIFICATION · ONSET, ANTITOXIN PRODUCT, ANTIBIOTICS

- **Onset.** The handout says symptoms appear within 12 hours. CDC describes foodborne botulism as typically appearing **12–36 hours** after toxin ingestion, and in rare cases as early as **6 hours** or as late as **10 days**.
- **"Trivalent antitoxin (types A, B and E)"** is what the handout teaches. The older anti-ABE trivalent product is described in CDC's guidelines as an older formulation; the product currently available in the United States is the **heptavalent antitoxin (types A–G)**.
- **Antibiotics** are not used for foodborne botulism, as the handout says; CDC states antimicrobials provide no benefit in treatment of botulism, and WHO notes antibiotics are not required **except in wound botulism**.

Diagnosis, treatment and prevention

Laboratory diagnosis

Diagnosis **depends on clinical manifestations**. The presence of **exotoxins in canned food** is detected by **ELISA**.

Treatment

Trivalent antitoxin serum (types A, B and E) is given IV, along with **respiratory support**. **Antibiotics are not used** to treat botulism.

Prevention

Swollen cans must be discarded.

STUDY AID

Quick comparison — incubation and key points

Compiled from the updated handout's own statements. Blank means the handout gives no figure.

Agent	Vehicle / source as stated	Incubation	Key features
S. typhimurium	Contaminated food; meat contaminated with rat excreta	**12–24 h**	Diarrhea, vomiting, fever for **2–5 days**; spontaneous cure usually
S. enteritidis	Contaminated food; **eggs** (can look perfectly normal)	**12–72 h**	Lasts **4–7 days**; most recover without antibiotics
Campylobacter	Food (specially poultry), milk, infected animals (e.g. dogs)	**1–7 days**	Cramps, nausea **but not vomiting**; mucus or blood in stool; GBS, reactive arthritis

Agent	Vehicle / source as stated	Incubation	Key features
Brucella	Unpasteurized milk products and meat; animal contact; aerosols	—	Undulating fever; urease **positive**; culture up to **4–6 weeks**
S. aureus	Food handler; carbohydrate-rich (cake, koshari) and protein-rich (milk) foods	**1–6 h**	**Preformed** heat-stable enterotoxin; vomiting, watery non-bloody diarrhea
B. cereus · **emetic**	Improperly stored, cooked rice	**1–6 h**	Preformed heat-stable enterotoxin; resembles *S. aureus*; < 24 h
B. cereus · **diarrheal**	Meat or vegetables	**8–16 h**	Heat-labile enterotoxin ↑ adenylate cyclase → ↑ cAMP; < 24 h
C. botulinum	Contaminated canned food	Handout: within **12 h** (see clarification)	Toxin blocks acetylcholine release → **flaccid paralysis**

E

FOR THE CAREFUL READER

Edition notes — what differs between the two sources

The 2026–2027 handout is the main line; the earlier YMnotes chapter (printed pages 106–114) supplies the notes and the supplement. Main topics are very similar.

Topic	Updated handout (main line)	Earlier edition
Learning objectives; Norovirus; *C. perfringens*; Enterobacteriaceae	Not present	Present — shown as earlier-edition notes (opener, section 1)
Salmonella reservoir; culture detail	Poultry as primary reservoirs; 37 °C for 24 h; pale yellow colonies; acid + gas except *S. typhi*; enrichment broths	Shorter wording; no reservoir statement
Intracellular survival of *Salmonella*	Not present	Present — shown as an earlier-edition note
Isolation timing	Stool, then urine, in the second week; blood culture in fluid media then subculture	Urine “after the second week”
Typhoid vaccines	“Four types … currently available”, listed a–d	“Several”, with b–d merged
***Campylobacter* diagnosis**	Dark-field illumination for darting motility	Culture on Skirrow’s stated as a line
***Brucella* culture**	Aerobic, 10% CO_2	“Aerobic” without CO_2
Botulism text	Same statements, including the 12-hour wording	Same — both are corrected by clinical clarifications
Other GIT pathogens	Not present	Leptospirosis, yellow fever, dengue, Ebola/Marburg — kept as the Supplement

Presentation changes only. Obvious spelling and typing slips were corrected (e.g. “Salmonellosis”, *melitensis*, “Hand-out”) without changing meaning, and lines the handout prints twice are given once. Unclear source lines are kept and flagged, not resolved from memory. Only clinical clarifications were checked against CDC/WHO.

SUPPLEMENT · EARLIER EDITION ONLY

Other GIT pathogens

This material follows botulism in the earlier YMnotes chapter (printed pages 113–114) and does **not** appear in the updated 2026–2027 handout. It is preserved for completeness. **Its presence here does not mean it is part of the current exam.**

Editorial: spelling corrected to *Aedes africanus* (printed "Aedes Africans") and *Aedes aegypti* (printed "Aedes egypti").

Leptospira and leptospirosis

Leptospira are **spirochetes** that live in stagnant water and lakes.

Morphology	**Spiral-shaped, motile**; one or both ends are hooked. Because they are so thin, live *Leptospira* are best observed by **dark-field microscopy**.
Culture	**Aerobic, at 30 °C**; need enriched media (**Stuart's media**) to enhance growth, which appears after **1–2 weeks**.

Leptospirosis (Weil's disease, 7-day fever) is a bacterial **zoonotic** disease caused by *Leptospira*. The infection is commonly transmitted to humans by allowing **fresh water contaminated by animal urine** to come in contact with **breaks in the skin, eyes or mucous membranes**. Occupational risk factors include **veterinarians, slaughterhouse workers, farmers, mine and sewer workers**. The disease is **not known to be spread from person to person**.

Pathogenesis: incubation period **4–14 days**. It is a **biphasic** disease; the patient is asymptomatic in between phases. The first phase is a variable febrile onset lasting **4–7 days**, with **hepatitis and jaundice**; the second phase is manifested by **meningitis (in 40%)** with intense headache and stiff neck. Hepatitis and nephritis may recur.

Laboratory diagnosis: organisms are present in **blood for the first 7 to 10 days**; after this period the microorganism can be found in **fresh urine** and kidney biopsy. **Direct examination:** dark-field microscope and PCR. **Culture:** colonies are identified by agglutination test and DNA probes. **Serologic diagnosis:** ELISA for IgM and total antibody. **There are no human vaccines.** Animal vaccines are effective for a few months.

Yellow fever (YF)

A **haemorrhagic fever** caused by the yellow fever virus, a **single-stranded enveloped RNA arbovirus** (arthropod-borne virus). The disease is present in **equatorial countries in Africa and South America** (Amazon basin). Man usually forms an **accidental, non-normal host** for the virus. **The liver and stomach are highly impacted.**

Pathogenesis and disease: incubation period **3–5 days**. Human activity can bring man into contact with mosquitoes, e.g. clearing of forests, agriculture. The virus multiplies in the **reticuloendothelial cells**, then viremia occurs and the virus is localized in **liver, spleen, kidneys and bone marrow**. This is a severe systemic disease in which **hemorrhages, and degeneration of the liver, kidney and heart** occur. **Jaundice** is a prominent feature due to liver necrosis. The **case-fatality rate can be 50%**.

1 · Jungle (sylvatic) yellow fever

The **monkey** is the main host; the virus is transmitted between monkeys by the *Aedes africanus* mosquito in Africa. Forest workers or visitors are infected accidentally and carry the infection to urban areas.

2 · Urban yellow fever

The disease is introduced into towns; the cycle is **man – mosquito – man**. The mosquito is *Aedes aegypti*.

Yellow fever	Earlier-edition text
Laboratory diagnosis	1 · **Intra-cerebral inoculation of mice** with a blood sample. 2 · Direct detection of the RNA virus by **PCR**. 3 · Detection of a **rising antibody titre** by neutralizing test, complement fixation, haemagglutination test.
Prevention and control	1 · Use of **insecticide** to kill mosquitoes and their eggs. 2 · **Vaccine: "17D strain"** live attenuated virus, given as a **single dose subcutaneously**. "Safe long-lasting protection is obtained for 10 years." **Vaccination is required for travelers to endemic equatorial zones.**

CLINICAL CLARIFICATION · YELLOW FEVER VACCINE

The earlier edition says protection lasts 10 years. **WHO now states that a single dose of yellow fever vaccine provides lifelong immunity and no booster is needed** (immunity develops in 80–100% of people within 10 days). The "10 years" wording reflects older practice.

Source: WHO, Yellow fever fact sheet (who.int).

Dengue, Ebola and Marburg

Dengue and Ebola all affect the liver and the gastrointestinal (GI) system, but with varying severity.

Dengue virus

RNA enveloped virus. The natural hosts are **monkeys and man**; **mosquitoes are the vector**. It has an **urban and a jungle cycle**. Symptoms include fever, headache, rash, **bone pain ("breakbone fever")**, myalgias and arthralgia. **Dengue hemorrhagic fever** is a deadly complication of dengue. The earlier chapter gives its case-fatality rate as 50%.

Ebola and Marburg viruses

RNA enveloped viruses that cause **fatal disease** in man: **haemorrhage, fever, diarrhea, vomiting and muscle pain. No known reservoir/vector hosts.** They are transmitted by **direct contact with infected body fluids**.

CLINICAL CLARIFICATION · DENGUE

A **50% case-fatality rate should not be read as the usual outcome of dengue hemorrhagic fever**. WHO's current fact sheet uses the term **"severe dengue"** and says early detection and access to proper medical care **greatly lower fatality rates**; WHO-affiliated sources put the case-fatality below 1% with appropriate care. Reported fatality in untreated severe dengue varies by source (roughly 10–20%). Learn the earlier edition's 50% only as that edition's statement.

Sources: WHO, Dengue and severe dengue fact sheet; WHO AFRO, Dengue; CDC, Severe Dengue 2015 case definition.

رَبِّ زِدْنِي عِلْمًا

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